Cloud-Based Restaurant Digital Menu Management System

Project Description

Project Title

Cloud-Based Restaurant Digital Menu Management System Using Microservices Architecture

Project Overview

Design and develop a cloud-based digital menu management system using **Microservices Architecture**. The system will work as a subscription-based SaaS (Software as a Service) platform for restaurants.

In this system, restaurants do not directly manage their menus. Instead, a central service provider (Admin) manages all restaurant menus. Whenever a restaurant wants to update or change its menu, it sends the updated menu design/image to the admin. The admin uploads the new menu image through the system, and the updated menu is automatically displayed on restaurant customer screens in real time.

The purpose of this project is to demonstrate:

- Cloud Computing
- Microservices Architecture
- Real-Time Data Synchronization
- REST API Development
- SaaS Application Design
- Distributed Systems

System Objectives

The system should:

- Allow restaurants to subscribe to the service
- Allow one central admin to manage all restaurant menus
- Display uploaded menus on customer-facing screens
- Automatically update menu screens whenever admin uploads a new menu
- Support multiple restaurant clients

Functional Requirements

1. Admin Panel

The system should contain one centralized admin panel.

Admin Features

The admin should be able to:

- Login securely
- View all subscribed restaurants
- Select a restaurant/customer ID
- Upload restaurant menu image/photo
- Replace/update menu image whenever restaurant requests changes
- View active menu currently displayed for each restaurant
- Manage restaurant subscription status

2. Restaurant Subscription Management

Restaurants subscribe to the menu service (manually).

Features:

- Each restaurant receives:
  - Unique Customer ID
  - Dedicated menu display screen
- Admin can:
  - Activate subscriptions
  - Deactivate subscriptions
  - View restaurant details

3. Digital Menu Display

Each restaurant should have customer-facing display screens/tablets/TVs that show:

- Uploaded menu image
- Latest active menu

The display should:

- Refresh automatically whenever menu is updated
- Show updates in real time
- Support full-screen mode

Customer Display Deployment

This is the MOST important missing part.

Students may not understand how screens work.

Add this section:

Customer Display Deployment

Each restaurant will have a dedicated display URL such as:

```
/display/REST001
```

This URL will be opened on:

- TV
- Tablet
- Monitor
- Browser

The screen should:

- Continuously display active menu image
- Listen for real-time updates
- Automatically refresh when admin uploads new menu

4. Real-Time Menu Updates

Whenever the admin uploads a new menu image:

- The updated menu should instantly appear on restaurant screens
- No manual refresh should be required

Suggested Microservices Architecture

The application should be divided into independent microservices.

1. Authentication Service

Responsibilities:

- Admin authentication
- JWT authorization
- Session management

Database:

- AdminUsers

2. Restaurant Management Service

Responsibilities:

- Manage restaurant/customer records

- Manage subscriptions
- Generate customer IDs

Database:

- Restaurants
- Subscriptions

3. Menu Upload Service

Responsibilities:

- Upload menu images
- Replace old menu images
- Store active menus
- Manage menu history (optional)

Database:

- RestaurantMenus

4. Real-Time Update Service

Responsibilities:

- Push updated menu images instantly to restaurant display screens
- Synchronize screens in real time

5. API Gateway

Responsibilities:

- Central request routing
- Authentication validation
- Communication between frontend and microservices

System Workflow

Restaurant Workflow

1. Restaurant subscribes to service
2. Restaurant receives customer ID
3. Restaurant sends updated menu image to admin
4. Admin logs into dashboard
5. Admin selects restaurant/customer ID
6. Admin uploads new menu image
7. System stores menu image in cloud storage
8. Real-time update service pushes updated menu to restaurant screens instantly

Customer Screen Workflow

1. Restaurant display screen loads active menu
2. System continuously listens for updates
3. When admin uploads new menu image, the screen updates automatically